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A Rational Theory of Mutual Funds' Attention Allocation

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1 Big picture

- **Question.** Do mutual fund managers add value by rationally allocating scarce attention across different kinds of information, and does that allocation change over the business cycle?
- **Core idea.** Skilled managers have limited information-processing capacity. They choose whether to learn more about aggregate shocks, which affect many stocks, or stock-specific shocks, which affect individual firms.
- **Main model result.** In recessions, managers optimally allocate more attention to aggregate shocks. In expansions, they allocate relatively more attention to stock-specific shocks.
- **Why recessions change attention.** Recessions have higher aggregate volatility and a higher price of risk. Aggregate shocks are more uncertain and more important for total portfolio risk, so learning about them is more valuable.
- **Empirical mapping.** Attention is not directly observed. The paper infers it from the covariance between fund portfolio weights and future aggregate or idiosyncratic payoff shocks.
- **Market timing prediction.** In recessions, fund portfolios should covary more with aggregate payoff shocks; this is interpreted as stronger market timing.
- **Stock picking prediction.** In expansions, fund portfolios should covary more with stock-specific payoff shocks; this is interpreted as stronger stock picking.
- **Dispersion prediction.** If skilled funds observe different private signals, recessions should increase cross-fund dispersion in portfolio strategies, betas, and returns.
- **Performance prediction.** Skilled funds should earn higher abnormal returns and higher alphas in recessions because information is more valuable when risk is more volatile and risk prices are high.
- **Data.** The empirical work uses CRSP mutual fund data, Thomson fund holdings, CRSP/Compustat stock data, industrial production shocks, recession indicators, and measures of risk prices.
- **Main empirical result.** The data support the predictions: attention shifts toward market timing in recessions, stock picking falls in recessions, dispersion rises, and fund performance improves in recessions.
- **Takeaway.** The paper provides a rational-inattention explanation for time-varying mutual fund skill and shows how business-cycle states discipline information-choice predictions.

2 Incremental contribution of the paper

2.1 Contribution relative to mutual fund performance research

- The mutual fund literature asks whether active managers have skill, but much of it documents average performance or cross-sectional performance persistence.
- This paper shifts the question from *whether* managers have skill to *when and what type* of skill is optimally deployed.
- It explains why market timing may be stronger in recessions and stock picking stronger in expansions, rather than treating fund skill as time invariant.
- It links portfolio holdings, not just returns, to managers' information choices. This makes the mechanism more directly testable.

Interpretation: The paper adds a state-contingent theory of active management. Skill is not a fixed scalar; it is an endogenous allocation of attention across information sources.

2.2 Contribution relative to rational inattention and information-choice theory

- Existing information-choice models with many assets are hard to solve because signals about aggregate and idiosyncratic risks generate nontrivial conditional correlations.
- The paper develops a tractable solution method using synthetic risk-factor assets. This allows investors to choose information about aggregate and stock-specific shocks in an N -asset equilibrium.
- The method generalizes prior two-asset or restricted-principal-component frameworks and creates testable predictions for portfolio holdings.
- The model is an equilibrium model, so for every skilled investor who outperforms, someone else must underperform; this is important for understanding average fund performance.

Interpretation: The technical contribution is not just solving another model. It creates a way to connect rational attention allocation to observable mutual fund portfolios and returns.

2.3 Contribution relative to business-cycle asset pricing

- The paper uses business-cycle variation in aggregate volatility and the price of risk as observable drivers of information choice.
- It shows that volatility and risk-price changes have separate effects: higher aggregate volatility makes aggregate information more valuable, and a higher price of risk makes learning about abundant aggregate risk even more valuable.
- It turns macro-finance state variables into predictions about micro-level portfolio choices by mutual funds.

Interpretation: The paper bridges asset pricing and delegated portfolio management: business-cycle risk changes not only prices, but also what information intermediaries rationally choose to process.

2.4 Contribution relative to empirical tests of fund skill

- The paper introduces fundamentals-based timing and picking measures rather than relying only on returns-based metrics.
- Fundamentals-based timing measures whether fund weights covary with future aggregate payoff shocks.
- Fundamentals-based picking measures whether fund weights covary with future stock-specific payoff shocks.
- This design helps rule out explanations based purely on sentiment, liquidity, or momentum because it links fund holdings to future fundamentals.

Interpretation: The incremental empirical value is to test information processing more directly: skilled managers should hold portfolios that predict future fundamental shocks.

3 Data and measurement

3.1 Mutual fund data

- The main sample is actively managed open-end U.S. equity mutual funds.
- The authors begin with the CRSP survivorship-bias-free mutual fund database, which provides returns and fund characteristics.
- They merge CRSP fund data with Thomson Financial holdings data to observe portfolios.
- The merged sample contains 3,477 funds.

- Fund controls include total net assets, age, expenses, turnover, flow, load, and flow volatility.

Interpretation: Holdings data are crucial because the model makes predictions about portfolio weights and their covariance with future shocks, not just about realized returns.

3.2 Stock-level and aggregate data

- Stock-level variables come from CRSP/Compustat.
- The aggregate market return is the value-weighted return of all CRSP stocks.
- Stock characteristics include market capitalization, book-to-market ratio, momentum, liquidity, standardized unexpected earnings, and leverage.
- Aggregate shocks are proxied using innovations in monthly seasonally adjusted industrial production.
- Recessions are measured using the NBER recession indicator.

Interpretation: Industrial production provides a fundamentals-based aggregate payoff shock. This is why the empirical timing measure is more closely connected to the model than a return-based market-timing measure.

3.3 Risk decomposition in individual stocks

- For each stock, the authors estimate a rolling CAPM regression using twelve months of data.
- The stock's aggregate risk is measured as $|\beta_t^m| \sigma_t^m$.
- The stock's idiosyncratic risk is measured as the residual standard deviation σ_t^ε .
- They compare aggregate risk, idiosyncratic risk, and the aggregate/idiosyncratic risk ratio across recessions and expansions.

Interpretation: This measurement step verifies the paper's motivating state variable: aggregate risk rises in recessions while idiosyncratic risk does not rise comparably.

3.4 Business-cycle price of risk

- The paper uses a measure of the price of risk based on yield spreads.
- Price of risk is defined as the difference between yields of Baa- and Aaa-rated U.S. corporate bonds.
- The price of risk is high in recessions and during periods of credit stress.

Interpretation: In the model, risk aversion ρ captures the price of risk. In the data, credit spreads proxy for that state variable.

3.5 Fundamentals-based market timing

The market-timing measure is

$$Ftiming_t^j = \frac{1}{TN^j} \sum_{i=1}^{N^j} \sum_{\tau=0}^{T-1} \left(w_{i,t+\tau}^j - w_{i,t+\tau}^m \right) (b_i z_{n,t+\tau+1}). \quad (1)$$

- $w_{i,t}^j$ is fund j 's weight in stock i .
- $w_{i,t}^m$ is the market portfolio weight in stock i .

- b_i is stock i 's loading on the aggregate shock.
- $z_{n,t+1}$ is the future aggregate payoff shock.

Interpretation: A fund has high market timing when it overweights assets that will load positively on future aggregate fundamentals and underweights assets that will load negatively.

3.6 Fundamentals-based stock picking

The stock-picking measure is

$$Fpicking_t^j = \frac{1}{N^j} \sum_{i=1}^{N^j} (w_{it}^j - w_{it}^m) z_{i,t+1}. \quad (2)$$

Interpretation: A fund has high stock-picking ability when it overweights stocks before positive stock-specific payoff shocks and underweights stocks before negative stock-specific shocks.

3.7 Dispersion measures

Portfolio dispersion is measured by

$$Portfolio\ Dispersion_t^j = \sum_{i=1}^{N^j} (w_{it}^j - w_{it}^m)^2. \quad (3)$$

Return dispersion is measured as the absolute deviation of a fund's abnormal return from the cross-sectional average abnormal return. Beta dispersion is constructed analogously using fund betas.

Interpretation: Dispersion is a key diagnostic for information processing. If managers observe heterogeneous signals, their portfolios and returns should differ more when information is especially valuable.

4 Model and theoretical mechanisms

4.1 Asset payoff structure

There is one riskless asset and n risky assets. For individual stocks $i \in \{1, \dots, n-1\}$,

$$f_i = \mu_i + b_i z_n + z_i, \quad (4)$$

where z_n is the aggregate shock and z_i is the stock-specific shock. The n th asset is a composite asset with payoff

$$f_n = \mu_n + z_n. \quad (5)$$

Interpretation: Each individual stock payoff has an aggregate component and an idiosyncratic component. The composite asset isolates aggregate risk.

4.2 Portfolio choice problem

Investors have mean-variance preferences over terminal wealth:

$$U_2^j = \rho E_j[W_j] - \frac{\rho^2}{2} V_j[W_j]. \quad (6)$$

The budget constraint in transformed risk-factor space is

$$W_j = rW_0 + \tilde{q}_j'(\tilde{f} - \tilde{p}r). \quad (7)$$

Optimal holdings are

$$\tilde{q}_j = \frac{1}{\rho} \hat{\Sigma}_j^{-1} \left(E_j[\tilde{f}] - \tilde{p}r \right). \quad (8)$$

Interpretation: Holdings rise with expected excess returns and with posterior precision. More precise signals allow managers to take larger positions.

4.3 Information choice and attention capacity

A skilled manager observes signals

$$\eta_j = z + \varepsilon_j, \quad (9)$$

where K_{ij} is the precision of manager j 's signal about shock i .

The capacity constraint is

$$\sum_{i=1}^n K_{ij} \leq K, \quad (10)$$

and the no-forgetting constraint is

$$K_{ij} \geq 0. \quad (11)$$

Interpretation: The manager's scarce resource is attention. She does not choose how much wealth to allocate to attention in the baseline; she chooses how to allocate a fixed amount of attention across shocks.

4.4 Equilibrium prices and learning from prices

Market clearing implies that transformed prices are linear in payoff shocks and supply shocks:

$$\tilde{p} = \frac{1}{r} (A + Bz + Cx). \quad (12)$$

Prices therefore reveal a public signal about payoff shocks:

$$\eta_p = B^{-1}(\tilde{p}r - A) = z + \varepsilon_p. \quad (13)$$

Interpretation: All investors learn from prices, but skilled investors also have private signals. This distinction is essential because price informativeness alone cannot create the cross-fund differences implied by heterogeneous private signals.

4.5 Proposition 1: volatility and attention

Proposition 1. For each skilled investor j , optimal attention allocation to risk i , K_{ij} , is weakly increasing in the variance σ_i of that risk.

Interpretation: Information is more valuable about more uncertain outcomes. Since aggregate volatility rises in recessions, skilled managers devote more attention to aggregate risk in recessions.

4.6 Proposition 2: price of risk and attention

Proposition 2. If the supply of risk i is sufficiently large, then the optimal attention allocation K_{ij} is weakly increasing in risk aversion ρ .

Interpretation: Learning about abundant aggregate risk reduces total portfolio uncertainty more effectively than learning about small idiosyncratic risks. When the price of risk is high, this risk-reduction channel becomes more valuable.

4.7 Propositions 3 and 4: dispersion

- Higher variance of an abundant risk increases dispersion in fund portfolios and returns.
- Higher risk aversion also increases portfolio and return dispersion when aggregate risk is abundant.

Interpretation: Passive portfolios would tend to comove more in recessions because payoffs are more aggregate. The model instead predicts higher dispersion because informed managers react to heterogeneous private signals.

4.8 Propositions 5–7: performance and alpha

- Higher aggregate variance increases skilled investors' excess returns.
- Higher risk aversion increases skilled investors' excess returns.
- CAPM alpha reflects information-processing capacity and rises in recessions when aggregate risk becomes more important.

Interpretation: Recessions are precisely the times when skill should become most visible in returns. This helps explain why average performance may look weak while state-contingent performance is strong.

4.9 Endogenous capacity extension

The baseline keeps total capacity fixed. An extension allows managers to choose capacity K at cost $C(K)$.

- The qualitative reallocation results remain unchanged.
- Managers choose more capacity in recessions because the marginal value of learning about aggregate risk rises.
- This amplifies dispersion and performance effects.

Interpretation: Endogenous capacity strengthens the main mechanism: recessions increase both the allocation of attention to aggregate risk and possibly the total amount of attention acquired.

5 Empirical predictions and identification

5.1 Prediction 1: cyclical attention allocation

- In recessions, portfolios should covary more with aggregate payoff shocks.
- In expansions, portfolios should covary more with stock-specific payoff shocks.
- Empirically, this means F_{timing} should rise in recessions and $F_{picking}$ should fall in recessions.

Interpretation: This is the most direct test of the rational attention mechanism because it uses portfolio weights and future fundamentals.

5.2 Prediction 2: dispersion

- Portfolio dispersion should rise in recessions.
- Beta dispersion should rise in recessions.
- Return dispersion should rise in recessions.

Interpretation: Dispersion distinguishes active information processing from passive exposure to common shocks. Passive portfolios should become less dispersed when common shocks dominate.

5.3 Prediction 3: performance

- Average abnormal returns should be higher in recessions.
- CAPM, three-factor, and four-factor alphas should be higher in recessions.
- The incremental effects of volatility and price of risk should be visible in performance regressions.

Interpretation: Performance is the payoff side of the same information-choice mechanism.

5.4 Identification logic

- The empirical strategy exploits cyclical variation in aggregate volatility and risk prices.
- Attention is inferred from covariance between fund holdings and future fundamentals.
- Fund fixed characteristics are controlled through age, size, expenses, turnover, flow, load, and flow volatility.
- Recession indicators identify state dependence; volatility and price-of-risk proxies separate the two model mechanisms.

Interpretation: The paper is not a randomized experiment. Identification comes from matching a set of theoretical predictions to multiple portfolio and return moments over the business cycle.

6 Tables and figures

6.1 Table I: individual stocks have more aggregate risk in recessions

Read:

- Panel A estimates the time-series relation between average stock risk measures and recession states.
- Aggregate risk is higher in recessions: the recession coefficient is positive and statistically meaningful in columns 1–2.
- Idiosyncratic risk is not significantly higher in recessions in the same way.
- The aggregate/idiosyncratic risk ratio rises in recessions.
- Panel B uses a large pooled stock-month sample and reaches the same broad conclusion.

Economic message: This table verifies the paper’s key state variable: recessions are periods in which aggregate risk becomes relatively more important than stock-specific risk.

	Aggregate Risk		Idiosyncratic Risk		Aggregate/Idiosyncratic Risk	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A: Time-Series Regression</i>						
Recession	1.348 (0.693)	1.308 (0.678)	0.058 (1.018)	0.016 (1.016)	0.098 (0.027)	0.097 (0.027)
MKTPREM		-4.034 (3.055)		-1.865 (3.043)		-0.215 (0.226)
SMB		8.110 (3.780)		12.045 (4.923)		0.167 (0.199)
HML		0.292 (5.458)		9.664 (8.150)		-0.308 (0.302)
UMD		-4.279 (2.349)		-1.112 (3.888)		-0.270 (0.178)
Constant	6.694 (0.204)	6.748 (0.212)	13.229 (0.286)	13.196 (0.276)	0.508 (0.013)	0.513 (0.014)
Observations	309	309	309	309	309	309
R-squared	6.85	9.70	0.10	3.33	8.58	10.52

(Continues)

indicator variable. The aggregate risk is twenty percent higher in recessions than it is in expansions (8.04% versus 6.69% per month), an economically and statistically significant difference. In contrast, the stock's idiosyncratic risk is not statistically different in expansions and in recessions. As a result, the ratio of aggregate to idiosyncratic risk increases from 0.508 in expansions to 0.606 in

Table I, Panel A: Time-series regression

	Aggregate Risk		Idiosyncratic Risk		Aggregate/Idiosyncratic Risk	
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel B: Pooled Regression</i>						
Recession	1.203 (0.242)	1.419 (0.238)	0.064 (0.493)	0.510 (0.580)	0.096 (0.021)	0.104 (0.024)
log(Size)		-0.145 (0.021)		-1.544 (0.037)		0.043 (0.002)
M Ratio		-0.934 (0.056)		-2.691 (0.086)		0.008 (0.004)
Momentum		0.097 (0.101)		2.059 (0.177)		-0.040 (0.005)
Leverage		-0.600 (0.074)		-1.006 (0.119)		-0.010 (0.003)
NASDAQ		0.600 (0.075)		1.937 (0.105)		-0.043 (0.005)
Constant	4.924 (0.092)	4.902 (0.095)	12.641 (0.122)	12.592 (0.144)	0.450 (0.009)	0.450 (0.009)
Observations	1,312,216	1,312,216	1,312,216	1,312,216	1,312,216	1,312,216
R-squared	0.62	2.90	0.000	19.33	0.58	7.56

^aFor each stock i and month t , we estimate a CAPM equation based on twelve months of data (a twelve-month rolling-window regression). This estimation delivers the stock's beta, β_{it} , and its residual standard deviation, σ_{it} . We use the stock's aggregate risk in month t as $|\beta_{it}\sigma_{it}^a|$ and its idiosyncratic risk as σ_{it}^a , where σ_{it}^a is the realized volatility in daily market return observations. Panel A reports results from a time-series regression of the average stock's aggregate risk, $\frac{1}{N} \sum_{i=1}^N |\beta_{it}\sigma_{it}^a|$, in columns 1 and 2, of the average idiosyncratic risk, $\frac{1}{N} \sum_{i=1}^N \sigma_{it}^a$, in columns 3 and 4, and of the ratio of aggregate to average idiosyncratic risk, in columns 5 and 6, on *Recession*. *Recession* is an indicator variable equal to 1 for every month the economy is in a recession according to the NBER, and zero otherwise. In columns 2, 4, and 6, we include several aggregate control variables: the market excess return (MKTPREM), the return on the small-minus-big portfolio (SMB), the return on the high-minus-low book-to-market portfolio (HML), the return on the up-minus-down momentum portfolio (UMD). The data are monthly from 1980 to 2005 (309 months). Standard errors (in parentheses) are corrected for autocorrelation and heteroscedasticity. Panel B reports results of pooled regressions of each stock's aggregate risk, $|\beta_{it}\sigma_{it}^a|$, in columns 1 and 2 and of its idiosyncratic risk, σ_{it}^a , in columns 3 and 4, and of the ratio of a stock's aggregate to idiosyncratic risk, in columns 5 and 6, on *Recession*. In columns 2, 4, and 6, we include several firm-specific control variables: the log market capitalization of the stock, *log(Size)*, the ratio of book equity to market equity, *B-M*, the average return over the past year, *Momentum*, the stock's ratio of book value to book debt plus book equity, *Leverage*, and an indicator variable, *NASDAQ*, equal to 1 if the stock is traded on NASDAQ. All control variables are lagged one month. The data are monthly and cover all stocks in the CRSP.

Table I, Panel B: Pooled regression

6.2 Table II: attention allocation is cyclical

Read:

- The dependent variables are fundamentals-based timing and fundamentals-based picking.
- The recession coefficient is positive for *Ftiming*, indicating more aggregate attention in recessions.
- The recession coefficient is negative for *Fpicking*, indicating less stock-specific attention in recessions.
- Price of risk and volatility measures also support the mechanism: higher risk prices and higher aggregate volatility tilt attention toward aggregate shocks.
- Controls include fund age, TNA, expenses, turnover, flow, load, and flow volatility.

Economic message: This is the most direct empirical evidence for the model's attention reallocation prediction: fund managers market-time more in recessions and stock-pick more in expansions.

6.3 Table III: portfolio and return dispersion rise in recessions

Read:

- Portfolio dispersion rises in recessions, with a recession coefficient of 0.204 in the simple specification and 0.118 with controls.
- Beta dispersion rises in recessions, supporting the mechanism that managers take different aggregate-risk positions.
- Return dispersion also rises in recessions.
- Volatility and price of risk help explain dispersion patterns, consistent with Propositions 3 and 4.

Economic message: The dispersion evidence supports an information-processing interpretation. Funds are not merely passively exposed to common shocks; they are actively taking heterogeneous positions based on different signals.

TABLE II
ATTENTION ALLOCATION IS CYCLICAL^a

	<i>Funding</i>			<i>Fpicking</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
Recession	0.011 (0.003)	0.012 (0.003)		-0.742 (0.138)	-0.680 (0.126)	
Price of Risk			0.019 (0.012)			-2.780 (0.514)
Volatility			0.003 (0.003)			-0.440 (0.123)
log(Age)		-0.001 (0.001)	-0.001 (0.001)	0.447 (0.061)	0.040 (0.144)	
log(TNA)		-0.001 (0.000)	-0.001 (0.000)	-0.130 (0.029)	-0.225 (0.052)	
Expenses		-0.208 (0.219)	-0.208 (0.219)	96.748 (11.200)	-90.819 (21.241)	
Turnover		-0.004 (0.001)	-0.004 (0.001)	-0.260 (0.063)	0.182 (0.087)	
Flow		-0.010 (0.011)	-0.010 (0.011)	0.637 (0.652)	1.305 (0.526)	
Load		0.006 (0.022)	0.006 (0.023)	-9.851 (1.951)	-9.876 (5.322)	
Flow Vol.		-0.006 (0.017)	-0.004 (0.017)	6.684 (1.042)	3.931 (1.164)	
Constant	-0.001 (0.001)	0.000 (0.002)	-0.001 (0.001)	3.082 (0.069)	3.238 (0.107)	3.119 (0.072)
Observations	221,488	221,488	221,488	165,029	165,029	165,029
R-squared	0.03	0.09	0.08	0.03	0.25	0.21

^aDependent variables: Fund *i*'s *Funding*_{*t*}^{*i*} is defined in equation (16), where the rolling window *T* is 12 months and the aggregate shock a_{t+1} is the change in industrial production growth between *t* and *t* + 1. A fund *i*'s *Fpicking*_{*t*}^{*i*} is defined as in equation (17), where i_{it+1} is the change in asset *i*'s earnings growth between *t* and *t* + 1. All are multiplied by 10,000 for readability. Independent variables: *Recession* is an indicator variable equal to 1 for every month the economy is in a recession according to the NBER, and zero otherwise. *log(Age)* is the natural logarithm of fund age in years. *log(TNA)* is the natural logarithm of a fund total net assets. *Expenses* is the fund expense ratio. *Turnover* is the fund turnover ratio. *Flow* is the percentage growth in a fund's new money. *Load* is the total fund load. *Flow Vol.* is the volatility of fund flows, measured from the last twelve months of fund flows. The last three control variables measure the style of a fund along the size, value, and momentum dimensions, calculated from the scores of the stocks in their portfolio in that month. They are omitted for brevity. All control variables are demeaned. *Flow* and *Turnover* are winsorized at the 1% level. *Price of Risk* is an indicator variable for periods with high default spread. Default spread is defined as a difference in yields of BBB and AAA-rated U.S. corporate bonds. Price of risk equals 1 if default spread is in the highest 10% of months in the sample. *Volatility* is an indicator variable for periods of high

TABLE III
PORTFOLIO AND RETURN DISPERSION RISE IN RECESSIONS^a

	Portfolio Dispersion		Beta Dispersion		Return Dispersion		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Recession	0.204 (0.027)	0.118 (0.025)	0.083 (0.015)	0.088 (0.014)	0.316 (0.147)	0.380 (0.146)	
Price of Risk							0.188 (0.088)
Volatility							0.637 (0.201)
log(Age)		0.210 (0.028)		-0.005 (0.002)		-0.121 (0.017)	-0.108 (0.018)
log(TNA)		-0.165 (0.014)		0.004 (0.001)		0.043 (0.009)	0.035 (0.010)
Expenses		31.986 (4.867)		4.162 (0.212)		28.330 (2.621)	25.526 (2.519)
Turnover		-0.113 (0.026)		0.013 (0.001)		0.090 (0.013)	0.076 (0.015)
Flow		-0.230 (0.108)		-0.004 (0.018)		-0.230 (0.223)	-0.280 (0.218)
Load		-1.658 (0.900)		-0.318 (0.041)		-4.071 (0.509)	-3.519 (0.517)
Flow Vol.		2.379 (0.304)		0.075 (0.027)		1.570 (0.240)	1.905 (0.242)
Constant	1.525 (0.024)	1.524 (0.022)	0.228 (0.006)	0.228 (0.006)	1.904 (0.084)	1.899 (0.077)	1.843 (0.078)
Observations	227,141	227,141	224,130	224,130	227,141	227,141	227,141
R-squared	0.10	4.80	1.35	8.10	0.19	7.00	7.83

^aDependent variables: Portfolio dispersion is the Herfindahl index of portfolio weights in stocks $i \in \{1, \dots, N\}$ in deviation from the market portfolio weights $\sum_{i=1}^N (w_{it}^i - w_{it}^m)^2 \times 100$. Return dispersion is $|return_t^i - return_t^m|$, where *return* denotes the (equally weighted) cross-sectional average. The CAPM beta comes from twelve-month rolling-window regressions of fund-level excess returns on excess market returns (and returns on SMB, HML, and MOM). Beta dispersion is constructed analogously to return dispersion. The right-hand-side variables, the sample period, and the standard error calculation are the same as in Table II.

The first dispersion measures we examine is *Portfolio Dispersion*, defined in

6.4 Table IV: fund performance improves in recessions

Read:

- Abnormal fund returns are higher in recessions.
- CAPM alpha is higher in recessions.
- Three-factor and four-factor alphas are also higher in recessions.
- The recession coefficients remain positive across specifications with fund controls.
- Price of risk and volatility also have positive effects in the richer specifications.

Economic message: This table links attention allocation to economic value. The same states in which aggregate attention rises are also states in which active funds earn higher abnormal returns and alphas.

TABLE IV
FUND PERFORMANCE IMPROVES IN RECESSIONS^a

	Abnormal Return		CAPM Alpha		3-Factor Alpha		4-Factor Alpha	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Recession	0.384 (0.056)	0.433 (0.059)	0.339 (0.048)	0.399 (0.050)	0.043 (0.034)	0.062 (0.026)	0.108 (0.041)	0.131 (0.033)
Price of Risk								0.052 (0.031)
Volatility								0.149 (0.061)
log(Age)		-0.015 (0.021)		-0.032 (0.008)		-0.023 (0.006)		-0.035 (0.006)
log(TNA)		0.023 (0.013)		0.040 (0.004)		0.018 (0.003)		0.019 (0.003)
Expenses		-5.120 (2.817)		-0.929 (0.892)		-5.793 (0.720)		-8.277 (1.248)
Turnover		0.021 (0.039)		-0.054 (0.010)		-0.087 (0.010)		-0.076 (0.009)
Flow		2.127 (0.672)		2.308 (0.172)		1.510 (0.096)		1.386 (0.056)
Load		-0.698 (0.457)		-0.810 (0.174)		-0.143 (0.129)		-0.371 (0.139)
Flow Vol.		-0.106 (0.588)		1.025 (0.137)		1.461 (0.109)		1.210 (0.104)
Constant	-0.032 (0.064)	-0.036 (0.063)	-0.060 (0.025)	-0.065 (0.024)	-0.059 (0.020)	-0.061 (0.018)	-0.051 (0.023)	-0.053 (0.021)
Observations	224,130	224,130	224,130	224,130	224,130	224,130	224,130	224,130
R-squared	0.01	0.57	1.15	10.70	0.03	6.20	0.16	5.50

^aDependent variables: *Abnormal Return* is the fund return minus the market return. The alpha come from twelve-month rolling-window regressions of fund-level excess returns on excess market returns for the CAPM alpha, additionally on the size (SMB) and the book-to-market (HML) factors for the three-factor alpha, and additionally on the momentum (MOM) factor for the four-factor alpha. The independent variables, the sample period, and the standard error calculations are the same as in Table II.

7 Short summary

- The paper develops a rational theory of mutual fund attention allocation.
- Skilled managers have fixed information-processing capacity and allocate it between aggregate and stock-specific shocks.
- In recessions, aggregate volatility and the price of risk rise.
- Higher aggregate volatility makes aggregate information more valuable.
- A higher price of risk further increases the value of learning about abundant aggregate risk.
- Managers therefore allocate more attention to aggregate shocks in recessions and more attention to stock-specific shocks in expansions.
- The paper measures attention using portfolio covariance with future aggregate and idiosyncratic fundamentals.
- The empirical results show that market timing increases in recessions and stock picking declines.
- Portfolio dispersion, beta dispersion, and return dispersion rise in recessions.
- Fund abnormal returns and alphas also rise in recessions.
- The evidence supports the idea that some mutual fund managers have skill and allocate attention rationally over the business cycle.

8 Conclusion

8.1 Core conclusion

Kacperczyk, Van Nieuwerburgh, and Veldkamp show that active mutual fund managers' information choices are state dependent. When aggregate risk and the price of risk are high, as in recessions, it is optimal to focus scarce attention on aggregate shocks. In expansions, when aggregate risk is less important, attention shifts toward stock-specific information.

8.2 Economic intuition

Information is valuable because it allows managers to resolve uncertainty and take better portfolio positions. The value of information depends on what risks matter most. In recessions, aggregate shocks affect many assets and command high risk premia, so learning about them improves portfolio choice. In expansions, stock-specific information becomes relatively more valuable.

8.3 Incremental takeaway

The paper's most important incremental contribution is to connect rational attention allocation, business-cycle risk, fund holdings, and fund performance in one framework. It explains why fund skill may appear different across booms and recessions, and it provides fundamentals-based empirical measures that directly test the model's information-choice mechanism.